

1 Package `nelder_mead`

1.1 Introduction to package `nelder_mead`

Package `nelder_mead` implements the Nelder-Mead minimization algorithm [1], also known as the polytope or amoeba method.

The Nelder-Mead algorithm is a derivative-free minimization algorithm; only evaluations of the objective function are required.

`nelder_mead` implements the "grid-restrained" Nelder-Mead algorithm published by A. Brmen et al. [2], and implemented in Common Lisp by Mario S. Mommer. Thanks to Andrej Vodopivec for the Maxima interface to the Common Lisp code.

References

[1] J.A. Nelder and R. Mead, "A simplex method for function minimization," The Computer Journal, vol. 7, pp. 308-313, 1965.

[2] A. Brmen, J. Puhan and T. Tuma, "Grid Restrained Nelder-Mead Algorithm", Computational Optimization and Applications, vol. 34, no. 3, pp. 359 - 375, 2006.

1.2 Definitions for package `nelder_mead`

`nelder_mead (obj, vars, init, nm_verbos = boolean, maxfuncalls [Function] = integer)`

Returns an approximate minimum of the objective function *obj*, as a function of the variables *vars*, starting at the initial point *init*.

`nm_verbos` tells `nelder_mead` to output additional information during the solution process. By default, it is `false`.

`maxfuncalls` tells `nelder_mead` to limit objective function evaluations to the number given. By default, there is no limit.

`nm_verbos` and `maxfuncalls` are optional, and may be omitted.

The objective function may be discontinuous, but if it is continuous, the algorithm ("grid-restrained" Nelder-Mead) is provably convergent. The objective function need not be differentiable; derivatives are not computed, not even approximately.

Examples:

```
(%i1) load ("nelder_mead") $
(%i2) nelder_mead (if x<0 then -x else x^2, [x], [4]);
(%o2) [x = 9.536387892694628E-11]
(%i3) f(x) := if x < 0 then -x else x^2 $
(%i4) nelder_mead (f, [x], [4]);
(%o4) [x = 9.536387892694628E-11]
(%i5) nelder_mead (x^4 + y^4 - 2*x*y - 4*x - 3*y, [x, y], [2, 2]);
(%o5) [x = 1.157212489168102, y = 1.099342680267472]
```

Appendix A Function and variable index

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